

近期 **3R** 模型及其应用路径综述

KYKT Dream

Internal research draft

ABSTRACT

近年来, 基于图像的三维重建从传统的匹配、位姿估计和多视图融合流程, 逐渐转向以可学习几何表示为核心的端到端或准端到端方法。DUS_t3R 系列及其后续工作将 pointmap、dense matching、视频深度、动态场景、长序列记忆和 pose-free reconstruction 等方向连接在一起, 形成了新的 3R 方法谱系。本文计划围绕模型关系、核心表示、适用场景、工程可复现性和应用落地路径, 对这一系列模型进行综述, 并总结本项目在调研、原型实现和实验规划中形成的经验。

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1 Introduction

近年来, 3D reconstruction 相关工作出现了明显的范式变化。传统流程通常依赖特征匹配、相机位姿估计、三角化和多视图融合; 近期 3R 模型则更多地把深度、匹配、位姿和点云表示纳入统一的可学习几何输出中。DUST3R 是这一变化中的代表性工作, 后续的 MAST3R、Fast3R、Spann3R、MonST3R、CUT3R、VGGT 等模型进一步扩展了匹配、速度、长序列、动态场景和统一视觉几何预测能力。

本文不把这些模型简单列成时间线, 而是从问题设置和能力分化出发, 讨论它们分别解决了哪些限制、引入了哪些新的系统代价, 以及在真实应用中如何组合使用。

1.1 Scope

本文重点覆盖以下方向:

- pointmap-based reconstruction and matching: DUST3R, MAST3R, MAST3R-SfM;
- scalable and multi-view reconstruction: Fast3R, MV-DUST3R+, VGGT;
- streaming and memory-based 3R: CUT3R, Spann3R, Point3R, SStream3R, LONG3R, LoGeR, Mem3R;
- dynamic and 4D reconstruction: MonST3R, POMATO, D²UST3R, Easi3R, RayMap3R;
- test-time verification and adaptation: Test3R, TTT3R, G-CUT3R;
- pose-free Gaussian and deployable 3D outputs: Splatt3R, InstantSplat, NoPoSplat, 3DGS-related methods;
- supporting priors and application links: Depth Anything, DINO, CoTracker, SAM-style masks, and KYKT-style result presentation.

1.2 Writing principle

The review should distinguish three kinds of statements:

- paper-level claims reported by the original authors;
- implementation or reproduction observations from local work;
- design interpretations and open hypotheses.

2 From classical reconstruction to learned pointmaps

TODO: explain the classical pipeline, why camera pose and matching are bottlenecks, and how DUST3R reframes the output as pointmaps.

3 Core branches of recent 3R models

3.1 Reconstruction and matching

TODO: DUST3R, MAST3R, MAST3R-SfM.

3.2 Scaling to more views

TODO: Fast3R, MV-DUST3R+, VGGT.

3.3 Streaming state and spatial memory

TODO: CUT3R, Spann3R, Point3R, SStream3R, LONG3R, LoGeR, Mem3R.

3.4 Dynamic scenes and 4D reconstruction

TODO: MonST3R, POMATO, D²UST3R, Easi3R, RayMap3R.

3.5 Test-time verification and correction

TODO: Test3R, TTT3R, G-CUT3R, geometry critic patterns.

3.6 Pose-free Gaussian and visual output

TODO: Splatt3R, InstantSplat, NoPoSplat, 3DGS output paths.

4 Comparative taxonomy

TODO: build tables by input assumption, output representation, temporal handling, memory mechanism, dynamic-scene support, model cost, and application readiness.

5 Application and deployment considerations

TODO: discuss how reconstruction outputs become usable artifacts: depth, point clouds, confidence maps, mesh/Gaussian outputs, visual inspection, report generation, failure-region marking, and integration with KYKT-style workflows.

6 Lessons from the Dream work

TODO: summarize methodological lessons from the long-running Dream research process: do not choose a single model prematurely; use evidence labels; separate flow validation from quality claims; design ablations before claiming architecture benefit; plan figures and deliverables around what has actually been verified.

7 Conclusion

TODO: write after the model inventory and paper reading are complete.

8 Bibliography